Piyush Hinduja

Salt Lake City, UT | (385) 507-6190 | piyushinduja20@gmail.com | Portfolio | LinkedIn

Education

Master of Science in Computer Science GPA: **3.91**/4.0
University of Utah, USA *2023 – 2025*

Bachelor of Engineering in Computer Engineering GPA: **3.85**/4.0
University of Mumbai, India *2019 – 2023*

Skills

Languages: *Expert (5+ yrs)*: Python, Java, JavaScript *Intermediate (3+ yrs)*: C, C++ *Beginner (1+ yrs)*: PHP, Kotlin, SQL, Shell
AI/ML: LLMs, PyTorch, HuggingFace, Langchain, TensorFlow, Scikit-Learn, EasyOCR, spaCy
Data Processing: RESTful APIs, Pandas, PySpark, Databricks, OCRs, Matplotlib, Seaborn, DuckDB, PowerBI, Tableau
Applications and Databases: FastAPI, Streamlit, Bash, Preswald, MongoDB, MySQL, Amazon RDS, Firebase, ChromaDB
Cloud & Miscellaneous: AWS (S3, SageMaker, Texteract, EC2), Azure, Docker, Kubernetes, Git, CI/CD, Agile/SCRUM, JIRA

Work Experience

AI Engineer (formerly Data Analyst) March 2024 – Present
University of Utah *Salt Lake City, Utah*

- Built staged **LLM-ETL pipelines** for targeted text extraction and insight generation across a large corpus of financial documents (from SEC.gov and EDGAR), enabling **90% faster**, context-aware information retrieval
- Performed deep NLP analysis using **encoder-only models like DeBERTa & FinBERT (PyTorch, Transformers)** for financial document processing, **improving the prediction accuracy from 66% to 94.13%**
- Developed a **Regex + Fuzzy Match system achieving 99.6% extraction accuracy**, ensuring precise entity identification and section-level text matching across heterogeneous filings
- Developed robust **prompt engineering** solutions to integrate LLMs & GenAIs (OpenAI, Anthropic, and Gemini) for fast and automated text analysis

AI Engineer July 2025 – Present
Giant Analytics *Remote*

- Fine-tuning **decoder-only models, like LLaMA 3.1**, on Indian Law data extracted via Indian Kanoon and Bharat Law APIs for domain-specific legal reasoning using **AWS SageMaker and S3**.
- Integrated the model into a **RAG-based SaaS platform** that retrieves and cites relevant cases during chat; projected to boost answer accuracy from **56% to 93%**, significantly **outperforming generic OpenAI & Gemini models**

Software Developer May 2025 – June 2025
Squeeze Media LLC *Orem, Utah*

- Built an AI-powered warm transfer system for inbound/outbound **Twilio calls using FastAPI, Deepgram SST, and WebSocket** streaming
- **Increased company sales by 28%** by ensuring more customers completed their calls without drop-offs or repeated queries

Full Stack Developer Intern June 2021 – July 2021
Exposys Data Labs *Bengaluru, India*

- Spearheaded the launch of a tourism project, integrating key features to **enhance user experience** and deliver valuable information about 50+ Indian travel destinations
- Utilized **MERN stack** for an efficient frontend and backend integration, creating a responsive, user-friendly interface and enabling content management

Projects

SEC Comment Letter Analysis | *LLMs, OCR, Claude API*

- Engineered a rolling-window algorithm to process and bundle SEC comment letter conversations, analyzing data from **50,000+ companies** via the EDGAR database
- Designed an **OCR pipeline using the EasyOCR library** to extract text from PDFs with **95% accuracy**, processing 2-page documents in under 2 seconds
- Constructed prompt-based labeling system using **Claude API** to classify letter content with **98.25% accuracy**, integrating metadata for contextual inference

Wireless Spectrum Analysis Research | *Python, Linux Bash, NVIDIA/CUDA*

- Engineered automated deployment system using Python and Linux Bash to streamline **NVIDIA/CUDA toolkit** integration on high-performance computing clusters, **reducing overhead by 90%**
- Analyzed wireless spectrum data to uncover scientific insights and **optimize power distribution across 20+ campus routing devices**

Road Damage Detection | *Object-Detection, YOLOv5, FasterRCNN* GitHub Link

- Collaborated with a team of four to develop a road damage detection system using **YOLOv5 and FasterRCNN**, integrating a user-friendly web interface (deployed via **Streamlit**), leveraging the RDD-2020 dataset
- Achieved model accuracies of **52.67% with YOLOv5 and 44.45% with FasterRCNN** by training and fine-tuning on different hyper-parameters
- The system plays a vital role, delivering a **70% improvement in road damage detection accuracy** over manual methods, helping lower the frequency of accidents and fatalities

Dependency Parser | *PyTorch, GloVe Embeddings* GitHub Link

- Architected an end-to-end **Transition-Based Dependency Parser in PyTorch**, implementing a multi-class classifier for predicting parsing actions based on stack and buffer states
- Designed modular pipeline incorporating data preprocessing, model architecture, training workflow, and evaluation metrics
- Integrated **GloVe word embeddings** for enhanced semantic representation, contributing to **UAS of 0.76 and LAS of 0.705**